

Pipes & Cisterns

COMPLETE LEARNING GUIDE

What You'll Learn

Pipes and cisterns problems are structurally identical to time and work problems, with one key extension: an outlet pipe or a leak contributes a NEGATIVE rate, since it removes from the tank rather than adding to it. The same LCM-based 'assign total capacity, compute each rate' method applies directly, but candidates frequently trip up by treating an emptying pipe's rate as positive, or by not checking whether pipes combined together can ever actually fill the tank at all — a net rate of zero is a genuine possibility, not a calculation error. This sub-module covers signed rates for inlets and outlets, combined filling and emptying, partial-time pipe operation, alternating pipe schedules, leak-adjusted filling time, and multi-pipe combinations — each with a complete worked numerical example.

1. Pipes as Signed Work Rates: Inlet vs Outlet

An inlet (filling) pipe contributes a POSITIVE rate toward the tank's capacity; an outlet (emptying) pipe or a leak contributes a NEGATIVE rate. Using the same LCM method as time and work, assign the tank a total capacity equal to the LCM of the given times, then compute each pipe's signed rate.

- Pipe A fills a tank in 6 hours, pipe B in 8 hours. Let capacity = $\text{LCM}(6,8) = 24$ units. A's rate = +4 units/hr, B's rate = +3 units/hr.

2. Two Filling Pipes Working Together

When two inlet pipes are opened together, their rates simply add, exactly as with combined workers in time-and-work problems.

- Using A (4 units/hr) and B (3 units/hr) from above, combined rate = 7 units/hr. Time to fill the 24-unit tank = $24/7$ "H 3.43 hours.

3. Inlet and Outlet Working Together: Net Rate

When an inlet and an outlet pipe are open simultaneously, the net rate is the inlet's rate MINUS the outlet's rate — if this net rate is positive, the tank fills; if negative, it empties; if exactly zero, the tank's level never changes at all.

- Pipe A fills a tank in 6 hours; pipe C (outlet) empties it in 12 hours. Capacity = $\text{LCM}(6,12) = 12$ units. A's rate = $+2/\text{hr}$, C's rate = $-1/\text{hr}$. Net rate = $+1/\text{hr}$. Time to fill = $12/1 = 12$ hours.

4. A Pipe Opened for Part of the Time, Then Another Joins

As with time-and-work, first compute the work done during the initial period at the initial rate, subtract from total capacity to get the remaining work, then divide by the new combined rate for the remaining time.

- Pipe A fills a tank in 6 hours, pipe B in 9 hours. A is opened alone for 2 hours, then B joins. Capacity = $\text{LCM}(6,9) = 18$ units. A's rate = $2/\text{hr}$, B's rate = $3/\text{hr}$. Work done by A alone in 2 hrs = 4 units. Remaining = $18 - 4 = 14$ units. Combined rate once B joins = $5/\text{hr}$. Additional time = $14/5 = 2.8$ hours. Total time = $2 + 2.8 = 4.8$ hours.

5. Alternate Opening of Pipes

When pipes are opened on alternate hours (or turns) rather than simultaneously, track cumulative work hour by hour (or cycle by cycle), exactly as in alternating-day time-and-work problems.

- Pipe A fills a tank in 6 hours, pipe B in 4 hours. Capacity = $\text{LCM}(6,4) = 12$ units. A's rate = $2/\text{hr}$, B's rate = $3/\text{hr}$. Opened on alternate hours starting with A: Hour 1 (A): 2, cumulative 2. Hour 2 (B): 3, cumulative 5. Hour 3 (A): 2, cumulative 7. Hour 4 (B): 3, cumulative 10. Hour 5 (A): 2, cumulative 12 — the tank fills exactly at the end of hour 5.

6. Cisterns With a Leak: Finding the Leak's Own Emptying Time

If a pipe's normal filling time is known and its ACTUAL filling time (with a leak present) is also given, the leak's own rate = (normal fill rate) $-$ (actual combined rate). Taking the reciprocal of that difference gives the time the leak alone would take to empty a full tank.

- A pipe can fill a tank in 8 hours, but due to a leak it actually takes 10 hours. Normal rate = $1/8$, actual combined rate = $1/10$. Leak's rate = $1/8 - 1/10 = (5 - 4)/40 = 1/40$. The leak alone would empty a full tank in 40 hours.

7. Multiple Inlet and Outlet Pipes Combined: The Zero-Net-Rate Trap

With several pipes open at once, sum all signed rates to get the net rate — but always check whether this net rate is actually positive before concluding the tank fills. A net rate of exactly zero means the tank's level never changes at all, regardless of how long the pipes stay open.

- Two inlet pipes fill a tank in 10 and 15 hours; one outlet pipe empties it in 6 hours. If all three are opened together, will the tank ever fill? Using $\text{LCM}(10,15,6) = 30$: rates = $+3/30$, $+2/30$, $-5/30$. Net rate = $(3+2-5)/30 = 0/30 = 0$. The tank never fills (or empties) — its level stays exactly where it started.

Common Pitfalls

- Treating an outlet or leak's rate as positive instead of negative when combining it with inlet pipes.
- Adding raw times of different pipes directly instead of converting each to a rate ($1/\text{time}$) first.
- Computing a leak's own emptying time as simply ($\text{actual time} \times \text{normal time}$) instead of correctly using the rate DIFFERENCE ($1/\text{normal} - 1/\text{actual}$).
- Assuming the tank always eventually fills whenever multiple pipes are open — a net rate of zero or negative means it never fills.
- In 'opened for part of the time, then joined' problems, forgetting to subtract the work already completed before dividing the remaining work by the new rate.
- Misreading a narrative word problem to determine which described pipe is the inlet and which is the outlet, especially when neither is explicitly labeled.
- In alternating-pipe problems, miscounting which pipe's turn falls on the final partial hour, causing the wrong rate to be used for the leftover work.

Key Takeaways

- Inlet pipes contribute positive rates; outlet pipes and leaks contribute negative rates — combined rate is the signed sum.
- Assign the tank a total capacity equal to the LCM of all given times to make every rate a whole number, exactly as in time and work.
- Leak's own emptying rate = ($\text{normal fill rate} - \text{actual combined fill rate}$); take the reciprocal for the leak's own time.
- A net rate of exactly zero means the tank neither fills nor empties — always check this before assuming the tank fills.
- For 'opened alone, then joined' problems, compute work already done first, then divide the REMAINING work by the new combined rate.
- Pipes and cisterns use the identical LCM-and-signed-rate method as time and work — recognizing this equivalence saves setup time.

Learn More

GeeksforGeeks — Pipes and Cistern Aptitude Questions

<https://www.geeksforgeeks.org/aptitude/pipes-and-cistern-questions-and-answers/>

IndiaBix — Pipes and Cisterns Aptitude Questions

<https://www.indiabix.com/aptitude/pipes-and-cistern/>